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Knowledge Management & Discovery Lab

Exercise 10

Additional Learning Sheet: Correlation Measures

1. χ^2 test for two categorical features.

This example follows the same workflow as in the lecture: build the contingency table, compute expected counts, compute the χ^2 statistic, then compare it to a critical value. The purpose is to check whether two categorical features look independent or correlated.

Two categorical features A and B were observed for 100 instances. The observed contingency table is shown in Table 1.

Table 1: Observed counts for features A and B .

A	B			Σ
	b_1	b_2	b_3	
a_1	30	10	20	60
a_2	10	25	5	40
Σ	40	35	25	100

The lecture formula is

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}, \quad e_{ij} = \frac{\text{count}(A = a_i) \cdot \text{count}(B = b_j)}{m}.$$

Here o_{ij} is the observed count and e_{ij} is the expected count under the independence assumption.

Step 1: State the null hypothesis.

$$H_0 : A \text{ and } B \text{ are independent.}$$

The alternative is that A and B are dependent, associated, or correlated.

Step 2: Compute the expected counts.

$$e_{ij} = \frac{\text{row total} \cdot \text{column total}}{m}.$$

For example,

$$e_{a_1, b_1} = \frac{60 \cdot 40}{100} = 24, \quad e_{a_1, b_2} = \frac{60 \cdot 35}{100} = 21.$$

Table 2: Expected counts under H_0 .

A	B			Σ
	b_1	b_2	b_3	
a_1	24	21	15	60
a_2	16	14	10	40
Σ	40	35	25	100

Step 3: Compute the χ^2 contributions.

Table 3: χ^2 calculation table.

Cell	o_{ij}	e_{ij}	$o_{ij} - e_{ij}$	$\frac{(o_{ij}-e_{ij})^2}{e_{ij}}$
(a_1, b_1)	30	24	6	$\frac{36}{24} = 1.50$
(a_1, b_2)	10	21	-11	$\frac{121}{21} = 5.76$
(a_1, b_3)	20	15	5	$\frac{25}{15} = 1.67$
(a_2, b_1)	10	16	-6	$\frac{36}{16} = 2.25$
(a_2, b_2)	25	14	11	$\frac{121}{14} = 8.64$
(a_2, b_3)	5	10	-5	$\frac{25}{10} = 2.50$
χ^2				22.32

Step 4: Compute the degrees of freedom. Feature A has $c = 2$ distinct values and feature B has $r = 3$ distinct values, so

$$df = (c - 1)(r - 1) = (2 - 1)(3 - 1) = 2.$$

Step 5: Compare with the critical value. At significance level $\alpha = 0.05$ and $df = 2$, use

$$\chi_{0.05,2}^2 = 5.99.$$

Since

$$22.32 > 5.99,$$

reject H_0 . The observed table is too different from the expected table under independence. Therefore, A and B are statistically correlated.

Interpretation for feature selection. If A and B are both input features, this may indicate redundancy: one feature may partly predict the other. If one of them is the target variable, then correlation is evidence that the other feature may be useful for prediction.

2. Practice: χ^2 test for a feature and a target variable.

The categorical feature Q has three values: low, medium, high. The target variable is *Response*, with values yes and no. The observed counts are shown in Table 4.

Table 4: Observed counts for feature Q and target variable $Response$.

Q	Response		$\sum$
	yes	no	
low	12	38	50
medium	25	25	50
high	38	12	50
$\sum$	75	75	150

Use the same formula:

$$e_{ij} = \frac{\text{row total} \cdot \text{column total}}{m}, \quad \chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}.$$

- State H_0 .
- Compute all expected counts.
- Compute all six χ^2 contributions.
- Compute $df = (c - 1)(r - 1)$.
- At $\alpha = 0.01$, use $\chi_{0.01,2}^2 = 9.21$. Decide whether Q and $Response$ are independent.
- Interpret the result for feature selection.

3. Pearson correlation for two numerical features.

Pearson correlation measures the degree of **linear** correlation between two numerical features. The value lies in

$$-1 \leq \rho(X, Y) \leq 1.$$

A value near +1 means strong positive linear correlation, a value near -1 means strong negative linear correlation, and a value near 0 means no detected linear correlation.

The formulas below are equivalent ways of writing Pearson correlation.

$$\rho(X, Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum_i (x_i - \bar{x})^2)(\sum_i (y_i - \bar{y})^2)}}.$$

The same coefficient can also be written as

$$r_{A,B} = \frac{\sum_{i=1}^m (a_i - \bar{A})(b_i - \bar{B})}{m\sigma_A\sigma_B} = \frac{\text{Cov}(A, B)}{\sigma_A\sigma_B}.$$

These formulas mean the same thing when covariance and standard deviation are computed with the same convention. The version with $m\sigma_A\sigma_B$ is the same as the square-root denominator version because

$$\sigma_A = \sqrt{\frac{1}{m} \sum_i (a_i - \bar{A})^2}, \quad \sigma_B = \sqrt{\frac{1}{m} \sum_i (b_i - \bar{B})^2}.$$

Two numerical features X and Y were measured for six instances.

Table 5: Numerical feature values for Pearson correlation.

Instance i	1	2	3	4	5	6
x_i	1	2	3	4	5	6
y_i	2	3	5	4	6	8

First compute the means:

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5, \quad \bar{y} = \frac{2 + 3 + 5 + 4 + 6 + 8}{6} = 4.67.$$

Table 6: Calculation table for Pearson correlation.

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
1	1	2	-2.5	-2.67	6.67	6.25	7.11
2	2	3	-1.5	-1.67	2.50	2.25	2.78
3	3	5	-0.5	0.33	-0.17	0.25	0.11
4	4	4	0.5	-0.67	-0.33	0.25	0.44
5	5	6	1.5	1.33	2.00	2.25	1.78
6	6	8	2.5	3.33	8.33	6.25	11.11
Σ					19.00	17.50	23.33

Therefore,

$$\rho(X, Y) = \frac{19.00}{\sqrt{17.50 \cdot 23.33}} = \frac{19.00}{20.21} = 0.94.$$

The correlation is strong and positive. Larger X values tend to occur together with larger Y values.

4. Practice: Pearson correlation.

Compute Pearson correlation for the following numerical features using the same $x_i - \bar{x}$ formula.

Table 7: Numerical feature values for Pearson practice.

Instance i	1	2	3	4	5	6
x_i	2	4	5	7	8	10
y_i	11	9	8	6	5	3

- Compute $\bar{x}$ and $\bar{y}$.
- Fill a calculation table with $x_i - \bar{x}$, $y_i - \bar{y}$, their product, and the two squared-deviation columns.
- Compute $\rho(X, Y)$.
- Interpret the sign and strength.

5. Spearman correlation for ordinal information.

Spearman correlation is Pearson correlation after replacing the original values by their ranks:

$$r_s = \rho(R(X), R(Y)) = \frac{\text{Cov}(R(X), R(Y))}{\sigma_{R(X)}\sigma_{R(Y)}}.$$

In this sheet, do **not** use the shortcut formula. Use the normal Pearson-on-ranks formula:

$$r_s = \frac{\sum_i (R(x_i) - \overline{R(X)})(R(y_i) - \overline{R(Y)})}{\sqrt{\left(\sum_i (R(x_i) - \overline{R(X)})^2\right) \left(\sum_i (R(y_i) - \overline{R(Y)})^2\right)}}.$$

Two variables X and Y were measured for eight instances. Rank 1 is assigned to the smallest value.

Table 8: Values and ranks for Spearman correlation.

Instance i	1	2	3	4	5	6	7	8
x_i	11	15	14	19	22	25	23	28
$R(x_i)$	1	3	2	4	5	7	6	8
y_i	10	17	13	18	21	24	20	30
$R(y_i)$	1	3	2	4	6	7	5	8

The mean rank for both variables is

$$\overline{R(X)} = \overline{R(Y)} = \frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8}{8} = 4.5.$$

Table 9: Calculation table for Spearman correlation using Pearson on ranks.

i	$R(x_i)$	$R(y_i)$	$R(x_i) - 4.5$	$R(y_i) - 4.5$	Product	Squares
1	1	1	-3.5	-3.5	12.25	12.25, 12.25
2	3	3	-1.5	-1.5	2.25	2.25, 2.25
3	2	2	-2.5	-2.5	6.25	6.25, 6.25
4	4	4	-0.5	-0.5	0.25	0.25, 0.25
5	5	6	0.5	1.5	0.75	0.25, 2.25
6	7	7	2.5	2.5	6.25	6.25, 6.25
7	6	5	1.5	0.5	0.75	2.25, 0.25
8	8	8	3.5	3.5	12.25	12.25, 12.25
Σ					41.00	42.00, 42.00

Hence,

$$r_s = \frac{41}{\sqrt{42 \cdot 42}} = \frac{41}{42} = 0.976.$$

The Spearman correlation is very strong and positive. The ordering induced by X is almost the same as the ordering induced by Y .

6. Practice: Spearman correlation without the shortcut.

Rank the following data and compute Spearman correlation as Pearson correlation on the ranks.

Table 10: Values for Spearman practice.

Instance i	1	2	3	4	5	6	7	8
x_i	5	8	10	12	15	18	20	22
y_i	21	19	18	15	13	11	8	7

- Compute $R(x_i)$ and $R(y_i)$, with rank 1 assigned to the smallest value.
- Compute $\overline{R(X)}$ and $\overline{R(Y)}$.
- Use the Pearson-on-ranks formula to compute r_s .
- Interpret the result.